Connes' Noncommutative Geometric Approach to RH: The Zeros as Absorption Spectrum

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Abstract

Connes [1] constructs a Hilbert space $L^2(\mathbb{A}_\mathbb{Q}/\mathbb{Q}^*)$ (square-integrable functions on the adele class space) and a self-adjoint operator D — the scaling operator on the adeles — whose *spectral action* reproduces the Weil explicit formula for $\zeta(s)$ as a Lefschetz trace formula. The Riemann zeros appear not as eigenvalues of D but as the *absorption spectrum*: frequencies at which the spectral density of D has a deficit. Making this precise requires a spectral interpretation of the explicit formula in which the zeros are genuine eigenvalues (and not just missing spectrum). We explain the adelic construction, its connection to the Weil conjectures over $\mathbb{F}_q$ (where the analogous trace formula is Grothendieck's), the Bost-Connes quantum statistical mechanical system, and the precise open problem: a suitable Sobolev space on the adele class space on which D has the Riemann zeros as eigenvalues.

1 The Adele Class Space

Definition 1 (Adeles and adele class space)**.** The *adele ring* of $\mathbb{Q}$ is

$$\mathbb{A}_\mathbb{Q} = \mathbb{R} \times \prod_p{}' \mathbb{Q}_p$$

(restricted product: $x \in \mathbb{A}_\mathbb{Q}$ has $x_p \in \mathbb{Z}_p$ for all but finitely many p). The multiplicative group of invertible adeles is $\mathbb{A}_\mathbb{Q}^*$. The *adele class space* is the quotient

$$\mathcal{X} = \mathbb{A}_\mathbb{Q}^*/\mathbb{Q}^*,$$

where $\mathbb{Q}^*$ embeds diagonally. The scaling action of $\mathbb{R}_+^*$ on $\mathcal{X}$ (multiplying all components) gives $\mathcal{X}$ a foliated structure that is fundamentally noncommutative.

Remark 2. The classical quotient $\mathbb{R}/\mathbb{Z}$ has nice topology; the quotient $\mathbb{A}^*/\mathbb{Q}^*$ does not: the orbit of any point under $\mathbb{Q}^*$ is dense. This is why $\mathcal{X}$ is studied via noncommutative geometry (C*-algebras, spectral triples) rather than classical topology.

2 The Scaling Operator and the Trace Formula

Definition 3 (The Hilbert space and operator)**.** Let $\mathcal{H} = L^2(\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^*, d^*x)$, where d^*x is the multiplicative Haar measure on $\mathbb{A}^*/\mathbb{Q}^*$. Define the *scaling operator*:

$$(U_\lambda f)(x) = f(\lambda^{-1}x), \quad \lambda \in \mathbb{R}_+^*.$$

Its generator $D = -i\frac{d}{d\log\lambda}$ is the self-adjoint operator of interest. The formal spectrum of D consists of real frequencies γ such that $e^{2\pi i\gamma}$ is an eigenvalue of some U_λ.

Theorem 4 (Connes [1]: Weil explicit formula as trace)**.** *Let h be a Schwartz-class function on $\mathbb{R}$ with Fourier transform $\hat{h}$. Then:*

$$\begin{aligned}\hat{h}(0) + \hat{h}(1) - \sum_{\rho:\,\zeta(\rho)=0} \hat{h}\left(\tfrac{\rho-1/2}{i}\right) \\ = \sum_p \sum_{m=1}^{\infty} \frac{\log p}{p^{m/2}}\left[h\left(\tfrac{m\log p}{2\pi}\right) + h\left(\tfrac{-m\log p}{2\pi}\right)\right] \\ + \hat{h}(0)\log\pi - 2\int_0^\infty h(t)\,\frac{e^{t/2}+e^{-t/2}}{e^t-1}\,\frac{dt}{2}\end{aligned} \tag{1}$$

is the Weil explicit formula, which Connes derives as the trace:

$$\mathrm{Tr}(h(D)|_{\mathcal{H}^\perp}) = \text{RHS of } (1),$$

where $\mathcal{H}^\perp$ is the orthocomplement of the trivial representation.

Remark 5 (Absorption spectrum, not eigenvalues). In (1), the zeros ρ appear with a *minus sign*: they create a deficit in the trace, reducing it from the "naive" value $\hat{h}(0) + \hat{h}(1)$. This is the *absorption spectrum*: the zeros are frequencies where D has *missing* spectral weight.

By contrast, the Hilbert-Pólya conjecture asks for the zeros to be *eigenvalues* of D — spectral contributions with a *plus sign*. These are different: a Hilbert space can have an absorption spectrum without those frequencies being eigenvalues (e.g., the continuous spectrum of a Schrödinger operator).

3 Comparison with the Function Field Case

Over a global function field $\mathbb{F}_q(C)$, the analogous construction works perfectly.

Theorem 6 (Grothendieck: trace formula over $\mathbb{F}_q$)**.** *For a smooth projective curve $C/\mathbb{F}_q$:*

$$\sum_{n=0}^{\infty} |C(\mathbb{F}_{q^n})|T^n = \frac{\det(1 - T\cdot\mathrm{Frob}|H^1)}{\det(1 - T\cdot\mathrm{Frob}|H^0)\cdot\det(1 - T\cdot\mathrm{Frob}|H^2)}.$$

The zeros of $Z(C,T)$ are the eigenvalues of Frob *on H^1; by Deligne's purity, $|\alpha_j| = q^{1/2}$. The Frobenius acts on $H^1_{\text{ét}}(C_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ — a finite-dimensional vector space — as a matrix, and its eigenvalues are the zeros with a plus sign.*

Proposition 7 (Comparison with Connes)**.**

Feature	*Function field* $\mathbb{F}_q(C)$	*Number field* $\mathbb{Q}$
Trace formula	*Grothendieck-Lefschetz*	*Weil explicit formula*
Operator	Frob *on* H^1 *(finite-dim.)*	D *on* $\mathcal{H}$ *(inf.-dim.)*
Zeros as	*Eigenvalues (plus sign)*	*Absorption (minus sign)*
Purity	*Proved (Deligne)*	*Open (RH)*
Spectral interpretation	*Complete*	*Partial*

Remark 8. The minus sign in Connes' formula is the key obstacle. Over $\mathbb{F}_q$, Frobenius has the zeros as eigenvalues, and purity (positivity of the form on H^1) forces $|\alpha_j| = q^{1/2}$. For D on $\mathcal{H}$, the zeros appear with a minus sign: the spectral action misses them, rather than hitting them. Connes' formulation is a trace formula for the *complement* of the zero locus.

4 The Bost-Connes System

The Bost-Connes system [2] is a quantum statistical mechanical system whose partition function is $\zeta(s)$ and whose "KMS states" (equilibrium states at inverse temperature β) detect the primes.

Definition 9 (Bost-Connes C*-algebra)**.** The Bost-Connes algebra $\mathcal{A}_{BC}$ is generated by:

- Elements $e(r)$ for $r \in \mathbb{Q}/\mathbb{Z}$ ("abstract roots of unity"), satisfying the relations of the group ring of $\mathbb{Q}/\mathbb{Z}$.
- Isometries μ_n for $n \in \mathbb{N}^*$, with $\mu_n^*\mu_n = 1$ and $\mu_n e(r)\mu_n^* = \frac{1}{n}\sum_{ns=r} e(s)$.

The time evolution is $\sigma_t(\mu_n) = n^{it}\mu_n$, $\sigma_t(e(r)) = e(r)$.

Theorem 10 (Bost-Connes [2])**.** *The partition function of the Bost-Connes system is* $Z(\beta) = \zeta(\beta)$*:*

$$Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \zeta(\beta), \quad \beta > 1.$$

The KMS$_\beta$ *states (Gibbs states at inverse temperature* β*):*

- *For* $\beta > 1$*: unique KMS state, factoring through evaluation at elements of* $\hat{\mathbb{Z}}^*$ *(profinite integers).*
- *For* $\beta = 1$ *(critical temperature): a family of KMS states parametrized by* $\mathbb{C}^*/\{\pm 1\}$*.*
- *For* $0 < \beta \leq 1$*: KMS states do not exist (above the critical point).*

The critical temperature $\beta = 1$ *corresponds to the pole of* $\zeta(s)$ *at* $s = 1$*.*

Remark 11 (Why $\beta = 1$ is critical). The Bost-Connes system undergoes a phase transition at $\beta = 1$ (equivalently, $s = 1$): below this temperature, the system is in a "disordered" phase (no KMS state); at this temperature, it breaks symmetry. The Riemann zeros ($s = 1/2 + i\gamma$) are on the critical line $\mathrm{Re}(s) = 1/2$, at the "halfway point" between $s = 0$ and $s = 1$. The Bost-Connes picture suggests the zeros are the resonant frequencies of the phase transition.

5 The Open Problem: Spectral Realization

Conjecture 12 (Spectral realization of zeros [1])**.** *There exists a Sobolev-type Hilbert space $\mathcal{H}_s \subset L^2(\mathbb{A}^*/\mathbb{Q}^*)$ on which the operator D is self-adjoint and has*

$$\operatorname{Spec}(D) = \left\{\gamma \in \mathbb{R} : \zeta\left(\tfrac{1}{2} + i\gamma\right) = 0\right\}.$$

Theorem 13 (Connes 1999: what is proved)**.**

1. *The Weil explicit formula* (1) *holds as an identity of distributions on $\mathbb{R}$.*
2. *The formula arises as a distributional trace of D on $\mathcal{H}$.*
3. *The zeros appear in the trace with a minus sign (absorption spectrum).*
4. *The Riemann Hypothesis is equivalent to the positivity of the Weil functional:*

$$\mathrm{RH} \iff \sum_\rho \hat{h}(\rho) \leq \hat{h}(0) + \hat{h}(1) - \text{local terms}$$

for all test functions $h \geq 0$ on $\mathbb{R}$.

Remark 14 (The gap: absorption vs. eigenvalues). Theorem 13(3) means the zeros appear in the trace formula with coefficient -1. For a classical operator with eigenvalues γ_j, $\operatorname{Tr}(h(D)) = \sum_j h(\gamma_j)$ (with a plus sign). Connes' absorption spectrum gives $-\sum_j h(\gamma_j)$ (minus sign).

To convert absorption to eigenvalues, one needs to:

1. Find a complementary Hilbert space $\mathcal{H}^c$ such that D has the zeros as eigenvalues on $\mathcal{H}^c$.
2. Show the trace on $\mathcal{H}^c$ gives the plus-sign formula.

This is Conjecture 12. It amounts to constructing the "cohomological complement" — the analogue of the ℓ-adic H^1 that Frobenius acts on over $\mathbb{F}_q$.

6 Connection to the Motives Program

Proposition 15 (Connes as motivic cohomology)**.** *The adele class space $\mathcal{X} = \mathbb{A}^*/\mathbb{Q}^*$ is the "absolute point" of Grothendieck's $\mathbb{F}_1$-geometry: the noncommutative geometry version of* $\operatorname{Spec}\mathbb{Z}$ *viewed as a curve over the field with one element. The Hilbert-Pólya operator H (if it exists on $\mathcal{H}^c$) would be the noncommutative-geometric analogue of Frobenius on the motivic cohomology $H^1(\operatorname{Spec}\mathbb{Z})$.*

Remark 16. This connects Connes' program to the program of [5]: the Weil conjectures use Frobenius on $H^1_{\text{ét}}(C)$ (over $\mathbb{F}_q$); Connes proposes that the Hilbert-Pólya operator is Frobenius on a noncommutative $H^1(\operatorname{Spec}\mathbb{Z})$. The two programs are attempting to construct the same missing cohomology from different directions.

Theorem 17 (Summary)**.** *Connes' noncommutative geometry approach to RH:*

1. *Constructs the adele class space $\mathcal{X}$ and scaling operator D.*

2. *Proves the Weil explicit formula as a trace formula on $\mathcal{X}$.*

3. *Shows the Riemann zeros appear as absorption spectrum (minus sign).*

4. *Reduces RH to positivity of the Weil functional (proved conditional on Conjecture 12).*

5. *The open step: Conjecture 12 — a Sobolev space on $\mathcal{X}$ on which D has the zeros as (positive, plus-sign) eigenvalues.*

The absorption-to-eigenvalue step is the arithmetic analogue of passing from the Weil explicit formula to the Grothendieck trace formula: over $\mathbb{F}_q$, this passage is achieved by étale cohomology; over $\mathbb{Q}$, it requires the missing arithmetic cohomology identified in [5, 6].

References

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